

ASSIGNMENT-VI

1. Write a C program to implement a **stack using an array** with menu-driven operations:

- `push()`
- `pop()`
- `peek()`
- `isEmpty()`
- `isFull()`

2. Implement a **stack using a singly linked list**.

Perform the following:

- Insert (push) elements
- Delete (pop) elements
- Display the stack

3. Implement a **queue using an array** with operations:

- `enqueue()`
- `dequeue()`
- `peek()`
- `isEmpty()`
- `isFull()`

4. Implement a **queue using a linked list** with:

- `enqueue()` (insert at rear)
- `dequeue()` (delete from front)
- Display function

5. Write a program to implement a circular queue using an array and demonstrate insertion and deletion.